PART A: WITHOUT LAGRANGE (Primal Method)

Q1: Identify support vectors, find hyperplane and margin for given dataset: (1,1,+1), (2,2,+1), (2,0,-1), (0,0,-1).

Q2: Find separating hyperplane for dataset: (3,3,+1), (4,3,+1), (1,1,-1), (2,1,-1).

Q3: Given hyperplane 2x1 + 2x2 - 4 = 0, find margin and identify support vectors among points (1,1), (2,2), (3,1), (0,0).

Q4: Choose correct maximum margin hyperplane for given dataset among options.

PART B: WITH LAGRANGE (Dual Method)

Q5: Formulate and solve dual problem for points (1,1,+1) and (-1,-1,-1).

Q6: Solve dual SVM for dataset: (2,2,+1), (4,4,+1), (2,0,-1), (0,0,-1).

Q7: Given alpha values, identify support vectors and compute hyperplane.

Q8: Use KKT conditions to identify support vectors.

Q9: Which points are support vectors?

Q10: If removing a point does not change hyperplane, is it a support vector?

Q1 Detailed Solution

Given points:
(1,1)+, (2,2)+, (2,0)-, (0,0)-

Step 1: Identify support vectors (closest points)
→ (1,1), (2,0), (0,0)

COMPUTE ALL DISTANCES : distance between every + and – point

(1,1) with (2,0) ==sqrt{(1-2)^2 + (1-0)^2} = sqrt{2}

(1,1) with (0,0)=sqrt{(1-0)^2 + (1-0)^2} = sqrt{2}

(2,2) with (2,0)=sqrt{0 + 4} =2

(2,2) with (0,0)=sqrt{4 + 4} = sqrt(8)

Minimum distance = **$\sqrt{2}$**

Occurs for:

- (1,1) & (2,0)
- (1,1) & (0,0)

Step 2: Use SVM conditions:
For +1: $w.x + b = 1$
For -1: $w.x + b = -1$

Equations:
(1,1): $w1 + w2 + b = 1$
(2,0): $2w1 + b = -1$
(0,0): $b = -1$

Step 3: Solve:
$b = -1$
$2w1 -1 = -1 \rightarrow w1 = 0$
$w2 -1 = 1 \rightarrow w2 = 2$

Step 4: Final:
$w = (0,2)$, $b = -1$
Hyperplane: $2x2 -1 = 0 \rightarrow x2 = 0.5$

Step 5: Margin:
$||w|| = 2 \rightarrow$ Margin $= 2/2 = 1$

Q5 Detailed Solution (Lagrange)
Points: (1,1)+, (-1,-1)-

Step 1: Dual constraint:
$\alpha1 = \alpha2 = \alpha$

Step 2: Weight:
$w = \Sigma\, \alpha i\, yi\, xi$
$= \alpha(1,1) - \alpha(-1,-1)$
$= \alpha(2,2)$

Step 3: Use condition:

$w.x + b = 1$
$(1,1)$: $2\alpha + 2\alpha + b = 1 \rightarrow 4\alpha + b = 1$

$(-1,-1)$: $-4\alpha + b = -1$

Step 4: Solve:
Add both:
$2b = 0 \rightarrow b = 0$
$4\alpha = 1 \rightarrow \alpha = 0.25$

Step 5: Final:
$w = (0.5,0.5)$, $b = 0$